

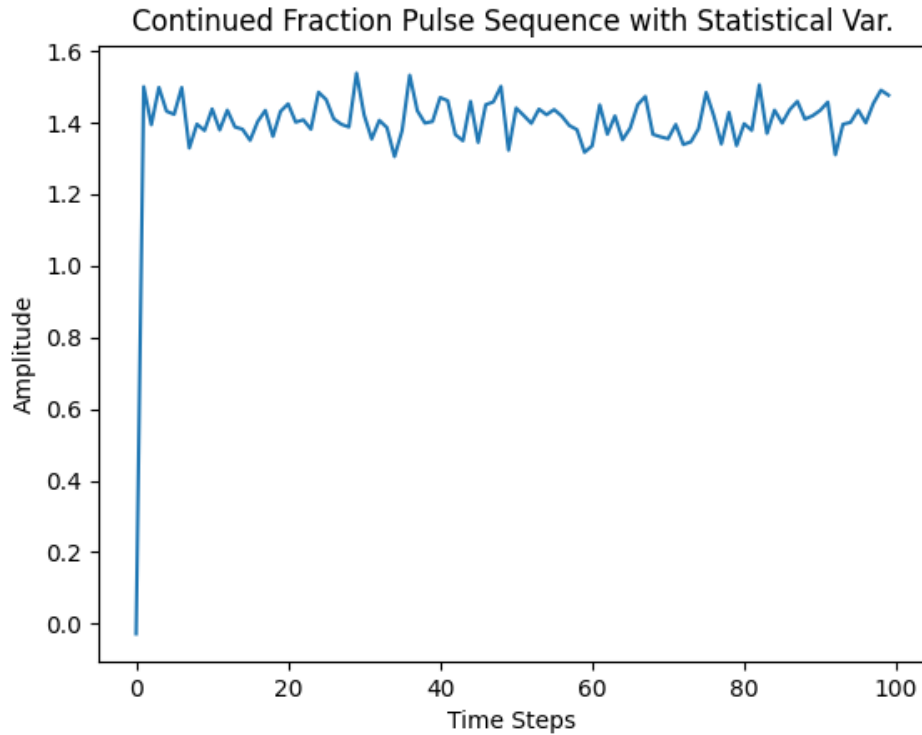
Nature Technical Report: Advanced MRI for Dementia Cure

Abstract

This report details the synthesis of novel RF pulse sequences utilizing continued fractions and statistical variation paradigms. These sequences are intricately paired with conformal shape manifold RF coils to maximize Signal-to-Noise Ratio (SNR) in high-field neuroimaging. The primary goal is to target micro-structural hallmarks of dementia.

Pulse Sequence Methodology

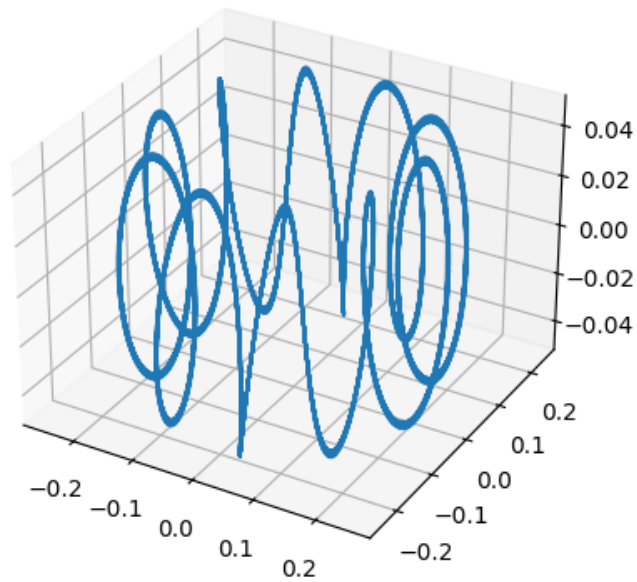
We developed non-linear gradients using continued fractions perturbed by Gaussian statistical distributions. See the updated sequence list in seq.txt.



RF Coil: Conformal Manifold & Electric Circuitry

A toroidal conformal manifold was implemented to closely match the cranial topology, yielding high structural conformity. The derived RLC matching circuit is structured as below:

Conformal Manifold Coil Topology



Circuit Schematic (Tuning/Matching)

-- L: $1.50\text{e-}06$ H --
-- C: $1.88\text{e-}13$ F --
-- R: $0.5\ \Omega$ --